HEMANT SURESH RATHOD

B.Tech Student - Artificial Intelligence & Data Science | AI, Full-Stack & Software Development

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PROFILE

Second-year B.Tech student in Artificial Intelligence & Data Science focused on building practical software with AI, agentic workflows, and modern web technologies. Hands-on experience developing full-stack applications, multi-agent research systems, AI-assisted products, and responsive interfaces. Strong interest in software engineering, applied AI, problem solving, and learning through real projects and hackathons.

EDUCATION

B.Tech in Artificial Intelligence & Data Science | NIAT / University Partner, Kolhapur, Maharashtra | *Expected 2029*

- **Second Year** - currently pursuing

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript

Frontend: React, Vite, React Router, Tailwind CSS, HTML5, CSS3

Backend & Data: Node.js, Express.js, MongoDB, Mongoose, Supabase, REST APIs

AI / Tools: LangChain, LangGraph, Google Gemini, Streamlit, ChromaDB, Sentence Transformers, LangSmith, Git, GitHub, Vercel

PROJECTS

Kestrel ResearchOS - Multi-Agent Research Assistant

- Built a 4-agent LangGraph workflow - Planner/Router, Retriever, Evidence Verifier, and Answer Synthesizer - for grounded research over technical documentation.
- Implemented local Sentence Transformers embeddings with ChromaDB retrieval, exact chunk citations, conflict detection, unsupported-query handling, and a Streamlit chat interface.
- Added an evaluation suite with 20 benchmark items; documented improved retrieval hit rate of 93.33%, citation precision of 100%, and answer faithfulness of 100%.

TravelMate+ - Maharashtra Travel Companion

- Developed a full-stack travel platform with React/TypeScript frontend and Node.js/Express backend backed by MongoDB.
- Implemented JWT authentication, trip planning, destination exploration, nearby-place search using browser geolocation, saved places, and persistent chat history.
- Integrated Google Gemini for a Maharashtra-focused travel assistant and itinerary planning, with English/Marathi localization and responsive UI.

PrimeTurf - Sports Turf Booking Platform

- Built a React web application for sports venue discovery and slot booking with separate user and admin experiences.
- Implemented authentication, booking flow, slot selection, receipts, booking history, admin controls, Supabase integration, and responsive UI.
- Focused on reusable React components, client-side routing, state/context management, and practical product-oriented UX.

LEADERSHIP & ACTIVITIES

- President, Entrepreneurship Club - supporting student-led initiatives, events, idea development, and entrepreneurial activities.
- President, Artificial Intelligence & Data Science Branch - representing students and contributing to branch-level coordination and activities.
- Participated in NxtWave x OpenAI Buildathon, Smart India Hackathon, NASA Hackathon, and UIDAI National Level Data Hackathon.

ACHIEVEMENTS & CERTIFICATIONS

- NxtWave x OpenAI Buildathon - Participation Certificate.
- NASA Hackathon - Participation Certificate.
- Smart India Hackathon - Participation / experience.
- UIDAI National Level Data Hackathon 2026 - Participant; project: Aadhaar Lifecycle Intelligence (ALI).

ADDITIONAL

- Interests: Applied AI, Agentic Systems, Full-Stack Development, Software Engineering, Hackathons, and Product Building.